

| Question Number | Answer                                                                                    | Additional Guidance        | Mark |
|-----------------|-------------------------------------------------------------------------------------------|----------------------------|------|
| 1(a)            | <ul style="list-style-type: none"> <li>• sclerenchyma (1)</li> <li>• xylem (1)</li> </ul> | Mark the first two answers | (2)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                          | Additional Guidance                                                    | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------|
| 1(b)(i)         | <p>An answer that includes one of the following points:</p> <ul style="list-style-type: none"> <li>• the {force / weight / strain / stress / tension / pressure} a {fibre / material} can take without breaking<br/>OR<br/>the {force / weight / strain / stress} required to make a fibre break (1)</li> </ul> | <p>IGNORE refs to cross sectional area</p> <p>ACCEPT snapping etc.</p> | (1)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Additional Guidance                                                                                                                                                                                                                                                                                                                                                | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(b)(ii)        | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• range of fibre lengths suggested / implied (1)</li> <li>• fibre variables taken into account (1)</li> <li>• appropriate environmental variable controlled (1)</li> <li>• appropriate description of how masses are added (1)</li> <li>• adding masses until fibre breaks / measure the mass that {breaks the fibre / that the fibre can hold before breaking} (1)</li> <li>• description of appropriate safety precaution with reason (1)</li> </ul> | <p>minimum two.<br/>ACCEPT if plot length vs mass added</p> <p>e.g. cross sectional area, diameter, width, plant type, age,<br/>ACCEPT if cross sectional area used in a calculation</p> <p>e.g. humidity, temperature, fibre pre-treatment<br/>NOT light intensity<br/>IGNORE same conditions</p> <p>ACCEPT carefully, small weights added one at a time etc.</p> | (4)  |

| Question Number        | Answer                                                                                                                                                                                                | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                                                 | Mark                   |                             |   |     |    |     |    |     |    |     |    |     |     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------|---|-----|----|-----|----|-----|----|-----|----|-----|-----|
| 1(c)(i)                | <p>table drawn with the following features:</p> <ul style="list-style-type: none"><li>• suitable table drawn (1)</li><li>• headings with units (1)</li><li>• all data correctly entered (1)</li></ul> | <table><tr><th>Fibre Length (/)<br/>mm</th><th>Tensile Strength (/)<br/>MPa</th></tr><tr><td>5</td><td>200</td></tr><tr><td>15</td><td>190</td></tr><tr><td>25</td><td>170</td></tr><tr><td>30</td><td>168</td></tr><tr><td>35</td><td>160</td></tr></table> <p>NOT mm<sup>-1</sup><br/>IGNORE order, both horizontally and vertically<br/>Lose mp2 if units in cells<br/>Inconsistent decimal places within a column loses mp3</p> | Fibre Length (/)<br>mm | Tensile Strength (/)<br>MPa | 5 | 200 | 15 | 190 | 25 | 170 | 30 | 168 | 35 | 160 | (3) |
| Fibre Length (/)<br>mm | Tensile Strength (/)<br>MPa                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |                             |   |     |    |     |    |     |    |     |    |     |     |
| 5                      | 200                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |                             |   |     |    |     |    |     |    |     |    |     |     |
| 15                     | 190                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |                             |   |     |    |     |    |     |    |     |    |     |     |
| 25                     | 170                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |                             |   |     |    |     |    |     |    |     |    |     |     |
| 30                     | 168                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |                             |   |     |    |     |    |     |    |     |    |     |     |
| 35                     | 160                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |                             |   |     |    |     |    |     |    |     |    |     |     |

| Question Number | Answer                                                                                                                                        | Additional Guidance                                                                                                                                          | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(c)(ii)        | <p>A calculation that includes the following steps:</p> <ul style="list-style-type: none"> <li>correct calculation of gradient (1)</li> </ul> | <p>e.g. <math>(160-200) \div 30 = -1.33</math> or <math>-1.3</math><br/>           Must have minus sign</p> <p>IGNORE <math>4/3</math> or other fraction</p> | (1)  |

| Question Number | Answer                                                                                                                                                                       | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(c)(iii)       | <p>A calculation that includes the following steps:</p> <ul style="list-style-type: none"> <li>find a value for c (1)</li> <li>correct assembly into equation (1)</li> </ul> | <p>ecf from 1cii</p> <p>e.g. <math>c = 200 - (-1.33 \times 5) = 206.7</math> (or <math>620/3</math>)<br/> or by extrapolation<br/> ACCEPT 206 to 207<br/> If minus not in 1cii do not penalise again.</p> <p>e.g. <math>y = -1.33x + 206.7</math> (206-207)<br/> ACCEPT <math>y = -4/3x + 620/3</math> as ecf<br/> ACCEPT mixture e.g. <math>y = -4/3x + 207</math><br/> Ecf their value of c from above</p> | (2)  |

| Question Number | Answer                                                                                                                             | Additional Guidance | Mark |
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| 1(d)(i)         | <ul style="list-style-type: none"> <li>as fibre length increases breaking strength decreases / negative correlation (1)</li> </ul> |                     | (1)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                      | Additional Guidance                                               | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|
| 1(d)(ii)        | <p>An answer that includes the following points.</p> <ul style="list-style-type: none"> <li>• bamboo may have different structure (to jute / coir) (1)</li> <li>• <b>number</b> of junctions may differ (1)</li> <li>• junctions may make fibre {weaker / stronger} at j (1)</li> <li>• {more / less} lignin (1)</li> </ul> | <p>ACCEPT description e.g. thicker<br/>But must be structural</p> | (3)  |